Thomas Le Corre | ATER

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Education

MSc in Mathematics of Randomness
Université Paris-Saclay *2020–2021*

Engineering degree with major in Mathematics and Data Sciences
CentraleSupélec *2017–2021*

BSc in Fundamental Physics
Université Paris-Sud *2017–2018*

Preparatory Classes for the MPSI/MP* competitive exams
CPGE Chateaubriand Rennes *2015–2017*

Professional Experience

ATER **ENS Lyon**
Unité de Mathématiques Pures et Appliquées *2025–2026*

PhD Student **ENS PSL / INRIA Paris**
Department of Computer Science, ARGO Team *2021–2025*

Final-year internship **EDF**
OSIRIS Department *2021*
Topic: Distributed control of diffuse flexibilities using Kullback–Leibler divergence under uncontrolled uncertainties

Gap-year internship **CEA Saclay**
Laboratoire des Sciences du Climat et de l'Environnement *2019*
Topic: Forced simulations of climate models.

PhD Thesis

Title: Distributed control of flexible loads on the power grid

Supervisor: Ana Bušić

Abstract: My PhD research focuses on providing new sources of flexibility that are crucial for integrating renewable energy into the power grid. The approach relies on modern smart technologies that enable automatic control of devices. The objective is to control a large population of devices in order to provide system services (load shaping or ancillary services) while preserving users' quality of service. The methodology combines techniques from controlled Markov processes, mean-field theory, and optimal transport.

Publications

The 2024 article establishes the theoretical framework of the first method studied in my PhD: mean-field control of piecewise deterministic Markov processes. The main theoretical results are stated and proved therein. The second key article is the one recently accepted at TMLR, introducing a formulation as an optimal transport problem with moment constraints. This constitutes the second method studied in my thesis. The remaining articles develop these two approaches and extend them to more applied settings relevant for power grid flexibility.

2026: **Moment Constrained Optimal Transport for Control Applications**, Thomas Le Corre, Ana Bušić, Sean Meyn, *Transactions on Machine Learning Research*.
https://openreview.net/forum?id=2hAtSpnat9

2025: **Mean-Field Control of Heterogeneous Piecewise Deterministic Markov Processes under Grid Constraints**, Thomas Le Corre, Ana Bušić, *SmartGridsComm 2025*.
https://ieeexplore.ieee.org/abstract/document/11204610

2025: **Moment Constrained Optimal Transport for Energy Demand Management of Heterogeneous Loads**, Julien Cardinal, Thomas Le Corre, Ana Bušić, *NETGCOOP 2025*.
https://link.springer.com/chapter/10.1007/978-3-032-09315-8_12

2024: **A decentralized algorithm for a mean field control problem of piecewise deterministic Markov processes**, Adrien Séguret, Thomas Le Corre, Nadia Oudjane, *ESAIM: Probability and Statistics*.
https://www.esaim-ps.org/articles/ps/pdf/2024/01/ps230042.pdf

Preprints

2025: **Mean Field Control of Thermostatically Controlled Loads as Piecewise Deterministic Markov Processes**, Thomas Le Corre, Adrien Séguret, Ana Bušić.
https://arxiv.org/abs/2511.01500

2025: **Moment Constrained Optimal Transport for Thermostatically Controlled Loads**, Thomas Le Corre, Julien Cardinal, Ana Bušić.
https://hal.science/view/index/docid/5226794

Teaching Experience

Below is a summary of my teaching activities. More detailed descriptions are provided on the following pages.

Position	Year	Institution	Audience	Level	Course	Hours (TD eq.)	Students	Format	Responsibilities
ATER	2025–2026	ENS Lyon	ENS Degree (Mathematics)	M1	Statistics	24h	12	Labs / Tutorials	Design of tutorials and lab sessions
ATER	2025–2026	ENS Lyon	CPES, 3rd year	L3	Introduction to Machine Learning	10h	9	Lectures / Tutorials	Design of lectures and tutorials
ATER	2025–2026	ENS Lyon	Agrégation candidates	M2	Modelling, Option A	27h (9h completed)	6	Lectures / Tutorials / Labs	Design of lectures, tutorials, and Python labs
ATER	2025–2026	ENS Lyon	CPES, 3rd year	L3	Biostatistics	36h	9	Lectures	Design of lectures, assignments, and exams
ATER	2025–2026	ENS Lyon	CPES, 3rd year	L3	Biostatistics	24h	9	Tutorials / Labs	Design of tutorials and R labs
Adjunct lecturer	2024–2025	ENS Paris	ENS degree (CS department)	M1	Models and Algorithms for Networks	15h	10	Tutorials / Labs	Design of tutorials, labs, and take-home assignments
Adjunct lecturer	2023–2024	Sorbonne Université	BSc Mathematics	L3	Differential Calculus and Optimization	18h	20	Tutorials	Weekly multiple-choice quizzes
Adjunct lecturer	2023–2024	Sorbonne Université	BSc Mathematics	L2	Linear and Bilinear Algebra I	36h	20	Tutorials	
Adjunct lecturer	2022–2023	Sorbonne Université	BSc Mathematics	L3	Differential Calculus and Optimization	36h	20	Tutorials	Weekly multiple-choice quizzes
Adjunct lecturer	2022–2023	Sorbonne Université	BSc Mathematics	L2	Linear and Bilinear Algebra I	36h	20	Tutorials	
Adjunct lecturer	2021–2022	Sorbonne Université	BSc Mathematics	L1	Mathematics for Scientific Studies II	36h	20	Tutorials	

Introduction to Machine Learning **ENS Lyon**

CPES, 3rd year *2025–2026*

This course follows the first-semester statistics course and aims to provide foundational knowledge in machine learning. It introduces several methods (linear regression, classification, dimensionality reduction, etc.) with applications in biology. I contribute to this course by delivering two 2-hour lectures and two 2-hour tutorial sessions. The course structure is already well established based on previous years, but I aim to incorporate additional biological examples to better motivate the mathematical concepts presented.

Statistics **ENS Lyon**

Master 1 *2025–2026*

This course provides an introduction to statistics for ENS students in mathematics and computer science. It covers estimators and their properties, various notions of estimator optimality, theoretical probability tools relevant to statistics, as well as statistical tests and their applications. I am responsible for the tutorial sessions and include additional numerical exercises for students who wish to explore computational aspects in greater depth.

Modelling, Option A **ENS Lyon**

Master 2 *2025–2026*

This course (27 tutorial-equivalent hours during the 2025–2026 academic year, 9 already completed) prepares students for the Modelling Option A (Probability and Statistics) exam of the French agrégation in mathematics. I am responsible in particular for the parts of the syllabus related to classical probability distributions, simulation methods for random variables, parametric tests, and goodness-of-fit tests. The course includes lectures (to be delivered), tutorials (3 hours completed), and laboratory sessions (6 hours completed). The labs are conducted in Python, which is the language used in the exam. The objective is to develop both student autonomy and efficiency, given the time constraints of the exam, as well as clarity in code and numerical result presentation. My background, more application-oriented than the traditional mathematics curriculum, allows me to provide a concrete approach to programming and modelling alongside theoretical material.

Biostatistics **ENS Lyon**

CPES, 3rd year (L3 level) *2025–2026*

This course is offered for the first time to CPES third-year students. I am therefore responsible for defining the pedagogical objectives in coordination with other instructors, as well as designing and delivering the course. I teach lectures, tutorials, and labs. The course content is partly inspired by Franck Picard's course for the standard ENS curriculum, but it has been expanded and deepened to fit a longer format. The syllabus covers descriptive and inferential statistics: estimators, hypothesis testing, linear models, and linear regression, with applications to life sciences. For the labs, students are beginners in R, and the goal is to enable them to use it in practice on biological datasets for data visualization and statistical inference (tests and linear regression in particular). The lab materials are available at: `https://github.com/tlecorre97/TP-CPES-3A-Biostatistiques`.

Models and Algorithms for Networks **ENS Paris**

Master 1 *2024–2025*

I was responsible for tutorials and labs for this course on network modelling, covering Markov processes and queueing theory. I designed the tutorials in collaboration with the professor in charge and also prepared and graded take-home assignments. While a set of exercises already existed, I contributed by writing full solutions for future use. The labs were conducted in Python, a language well known to ENS computer science students, and focused on implementing network algorithms (notably using the `networkx` library) and modelling queueing systems. One of the labs is available at: `https://colab.research.google.com/drive/1HAAxVixX_Li3ra6QcWs4g-2_cwi5dLWu?usp=sharing`.

Differential Calculus and Optimization **Sorbonne Université**

BSc Mathematics (3rd year) *2022–2024*

I taught tutorials for two years in this course, which covers differentiability in real vector spaces, convexity, constrained continuous optimization, and numerical methods. Each tutorial began with a 10-minute multiple-choice quiz to assess students' understanding of the previous week's material. Although time-consuming to design and grade, this proved very helpful for students and encouraged regular revision. The pedagogical approach avoided full board corrections; instead, individualized guidance was provided based on each student's progress. While this limited the total amount of material covered, it fostered deeper engagement and more personalized feedback.

Linear and Bilinear Algebra I **Sorbonne Université**

BSc Mathematics (2nd year) *2022–2024*

I taught tutorials and labs for this course introducing the fundamentals of algebra: vector spaces, bases and dimension, linear maps, matrices, bilinear and quadratic forms, and diagonalization.

Mathematics for Scientific Studies II **Sorbonne Université**

BSc Mathematics (1st year) *2021–2022*

I taught tutorials and labs for this course covering matrix calculus and an introduction to probability theory: counting, random variables, etc.